TELE6530 Lab Module #3

System Performance Monitoring UML

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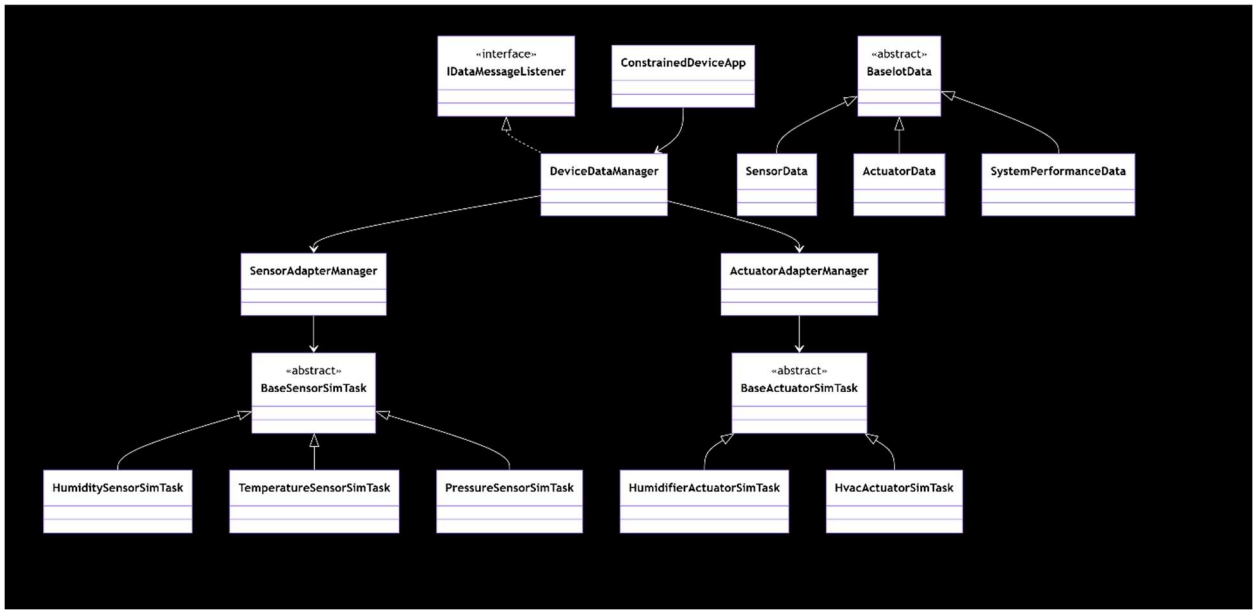
Destination for the submission: Canvas/Assignments/Lab03

Your development system's operating environment: Ubuntu 24

Markdown Documentation

This implementation builds a complete simulation and management infrastructure for a Constrained Device Application (CDA) that generates realistic environmental sensor telemetry (temperature, humidity, pressure), processes actuation commands for HVAC and humidifier devices, and performs on-device analytics such as triggering HVAC based on temperature thresholds. The system uses a layered architecture with data containers, simulation tasks, adapter managers, and a central Device Data Manager that orchestrates all components through callback mechanisms.

UML DIAGRAM


«interface»
IDataMessageListener
ConstrainedDeviceApp
«abstract»
BaseIotData
DeviceDataManager
SensorData
ActuatorData
SystemPerformanceData
SensorAdapterManager
ActuatorAdapterManager
«abstract»
BaseSensorSimTask
«abstract»
BaseActuatorSimTask
HumiditySensorSimTask
TemperatureSensorSimTask
PressureSensorSimTask
HumidifierActuatorSimTask
HvacActuatorSimTask